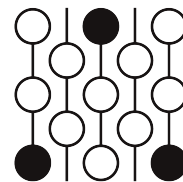


Review of Submission 1A

First Proof Second Batch

Reviewer 1

June 4–8, 2026



1 Review

Correctness. As far as I can tell, the solution is completely correct in all its technical details. The only issue that I noticed, which is not serious, is that in the middle of the solution there is a high-level description of the strategy to complete the proof which is at best misleading and at worst simply a false description of what’s going on. However, this statement does not play a technical role in the proof and, as I stated, all the technical details are correct. This is further discussed in an inline comment below.

Novelty. I am not aware of any proof that closely resembles appearing in the literature. However, I am not confident that no such proof exists and nothing about this solution is sufficiently surprising to rule that out.

It is interesting to note that this solution is extremely similar, in both the overarching ideas and in the specific way that those ideas are carried out, to the solution written by the authors of the problem. The main difference between the two solutions, which is mostly superficial, is that while this solution uses an infinite dimensional $\mathbb{F}_2$ vector space (which is identified with the set of finite subsets of $\mathbb{N}$ with the operation of symmetric difference), the solution by the problem’s authors essentially uses a free infinite dimensional $\mathbb{Z}$ -module (which is identified with the set of nonzero rational numbers with the operation of multiplication). However, this difference plays almost no role in the solutions.

This solution also has some similarity to two of the other AI solutions to this problem as well as to a technique introduced by Dan Turetsky (in the paper “Coding in the automorphism group of a countable structure”). However, it feels to me that some general features will be shared by any solution to this problem, so perhaps this is not too surprising.

Exposition. Overall, the exposition is reasonably clear and straightforward, though there are a number of minor points that could be improved upon. For example, the solution often switches between referring to a certain set as V and as S . In context, there are valid reasons to introduce both V and S , but once they are introduced I think the proof would be easier to follow if one of the two were used consistently. The discussion of “equality classes” in the final few paragraphs of the proof also seems needlessly confusing. However, in spite of these issues, the general quality of the exposition is well within what I would consider the acceptable range for published papers.

2 Recommendation

Essentially flawless. There is one misleading or false high level description of part of the proof strategy, but no discernible technical errors.

3 Submission with inline comments

A computably AUT-countable example with no finite existential automorphism basis

Theorem 1. *There is a computably represented countable structure $\mathcal{A}$ such that every automorphism of every copy of $\mathcal{A}$ is computable from that copy, and hence $\mathcal{A}$ is computably AUT-countable on a cone, but for every finite parameter set $P \subseteq A$ there is an automorphism π of $\mathcal{A}$ whose graph is not Σ_1^{in} -definable over $P \cup \pi(P)$.*

Proof. Let $V = [\omega]^{<\omega}$, with addition $+$ given by symmetric difference; thus V is a countably infinite vector space over $\mathbb{F}_2$. Fix a computable sequence $(D_i)_{i \in \omega}$ of finite subsets of V such that every finite subset of V occurs cofinally often.

We use a one-sorted computable coding of two sorts. The domain is the computable disjoint union

$$(\{0\} \times \omega) \sqcup (\{1\} \times V),$$

identified effectively with ω . We write its two unary predicates as I and S . The language also has a binary relation $<$ and a ternary relation R . On I the relation $<$ is the usual order of ω , and $<$ is false if one of its arguments is outside I . For $i \in I$ and $x, y \in S$, put

$$R(i, x, y) \iff x + y \in D_i,$$

where we identify elements of the S -sort with vectors in V ; on all other triples R is false. This is a computable relational structure.

First compute its automorphism group. Since $(I, <)$ has order type ω , every automorphism fixes I pointwise. If $f \in \text{Aut}(\mathcal{A})$, let $c = f(0_V)$. For every $h \in V$ choose i with $D_i = \{h\}$. Then $R(i, x, x + h)$ for all $x \in V$, and preservation of R gives $f(x + h) = f(x) + h$. Taking $x = 0_V$ gives

$$f(h) = c + h \quad (h \in V).$$

Conversely, for each $c \in V$, the map τ_c which fixes I pointwise and sends $x \in S$ to $x + c$ preserves all differences $x + y$, hence preserves R . Thus

$$\text{Aut}(\mathcal{A}) = \{\tau_c : c \in V\}.$$

This already implies computable AUT-countability, with no cone oracle. Let $\mathcal{B} \cong \mathcal{A}$, and let $\sigma \in \text{Aut}(\mathcal{B})$. Transporting σ to $\mathcal{A}$, it is some translation τ_c . Choose an element i of the I -sort of $\mathcal{B}$ corresponding to some index with $D_i = \{c\}$. A Turing program with oracle the atomic diagram of $\mathcal{B}$ and with this natural number i hardwired computes σ : it queries the sort predicate, outputs n

on inputs n in the I -sort, and on inputs x in the S -sort it searches for the unique y with $R(i, x, y)$. Hence every automorphism of every copy is computable relative to the copy.

It remains to prove the promised failure of Σ_1^{in} -definability. Fix a finite parameter set $P \subseteq A$. Let

$$M = \max(P \cap I)$$

if $P \cap I \neq \emptyset$, and put $M = -1$ otherwise. Let

$$W = \text{span}\left(\bigcup_{i \leq M} D_i\right) \leq V$$

(where $W = \{0\}$ if $M = -1$). The subspace W is finite. Choose a nonzero $h \in V \setminus W$, and let $\pi = \tau_h$. Put $Q = P \cup \pi(P)$. Since the quotient V/W is infinite and $Q \cap S$ is finite, choose $x \in V$ such that the two cosets

$$x + W, \quad x + h + W$$

are disjoint from $Q \cap S$. Let $y = x + h$.

We shall show that the graph-pair (x, y) is not isolated inside $\text{graph}(\pi)$ by any existential formula over Q . This suffices, because if a countable disjunction of existential formulas defined exactly the graph, then any disjunct true of (x, y) would itself have to be contained in the graph.

Reviewer 1

It's not immediately obvious what is meant by the phrase "the graph-pair (x, y) is not isolated inside $\text{graph}(\pi)$ by an existential formula over Q ". The most reasonable interpretation is that there is no existential formula φ (with parameters over Q) such that $\varphi(x, y)$ holds and there is *no other* pair (x', y') in the graph of π for which $\varphi(x', y')$ holds. Under this interpretation, this is neither necessary or sufficient to finish the proof. However, the rest of the solution (which does correctly finish the proof) makes it clear that the intended interpretation is that for any existential formula φ (with parameters over Q) for which $\varphi(x, y)$ holds, there is also some pair (x', y') which is *not* in the graph of π for which $\varphi(x', y')$ holds. I think this is at best a highly misleading way to phrase this idea, but since the rest of the solution is correct and makes clear what was meant, this is not a serious issue.

Let

$$\exists \bar{z} \theta(\bar{q}, u, v, \bar{z})$$

be an existential formula with parameters $\bar{q}$ from Q , and suppose it is true at $(u, v) = (x, y)$. Choose witnesses. Passing to one true conjunction of literals in a disjunctive normal form of the quantifier free part, it is enough to preserve the truth values of the finitely many literals in that conjunction.

Call an I -sort value *low* if it is at most M , and *high* otherwise. All low markers have their associated finite sets D_i contained in W . Let F be the finite set of all S -sort elements occurring among x, y , the parameters, and the chosen witnesses, and put $C = y + W$. We shall add a vector t precisely to those members of F which lie in C . The coset C contains no parameter and does not contain x .

Choose $t \in V$ outside the following finite forbidden set. First avoid $t = 0$ and all choices causing a shifted element of $F \cap C$ to collide with an unshifted element of $F \setminus C$. Also, for every $i \leq M$, every $a \in F \cap C$, every $b \in F \setminus C$, and every $d \in D_i$, avoid $t = a + b + d$. These exclusions

guarantee that all equalities and all R -literals whose first argument is a low marker are preserved: within C all differences are unchanged; outside C nothing is changed; and across the cut the original difference is not in W , while the new difference is kept out of each low D_i .

There are also finitely many high I -sort equality classes among the chosen witnesses. For each such class μ , and for each pair consisting of a positive and a negative R -literal with first argument in μ and other two arguments in the S -sort, we additionally require that the two resulting differences after the vector shift be unequal. If the two differences are shifted in the same way this inequality already follows from the original true diagram; otherwise it excludes one further value of t . Thus only finitely many more values are forbidden, and we choose t avoiding all of them.

Now perform the vector shift by this t . For each high equality class μ , let A_μ be the finite set of new differences required by the positive R -literals with first argument in μ , and let B_μ be the finite set of new differences forbidden by the negative such literals. By the choice of t , $A_\mu \cap B_\mu = \emptyset$. Replace the high I -sort witnesses by new indices, all above M and in the same equality and order pattern as before, with associated sets exactly A_μ . This is possible because each finite subset of V occurs cofinally often among the D_i 's. Sort predicates, equalities, and $<$ -literals are preserved by construction. Literals involving R with an argument in the wrong sort remain false, low-marker R -literals were handled above, and high-marker R -literals are made true or false according as their resulting difference lies in A_μ or not.

Consequently the same existential formula is true of the new pair

$$(x, y + t) = (x, x + h + t).$$

But $t \neq 0$, so $y + t \neq y = \pi(x)$. Hence this new pair is not in the graph of π . Therefore no existential formula over $Q = P \cup \pi(P)$ which is true of $(x, \pi(x))$ can be contained in the graph of π , and the graph of π is not Σ_1^{in} -definable over Q .

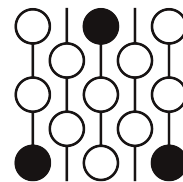
Since P was arbitrary, the required property holds for every finite parameter set. $\square$

Review of Submission 1A

First Proof Second Batch

Reviewer 2

June 4–8, 2026



1 Review

Correctness. The argument is correct. There are some statements that are hard to understand due to their wording, but they can generally be understood with some effort. The level of details provided is in line with standard mathematical practice, i.e., some small steps need to be easily verified by the reader.

Novelty. This argument is the closest to the original human solution. It still uses the idea of affine $\mathbb{F}_2$ -space which also appeared in *Coding in the automorphism group of a computably categorical structure* (Turetsky), *Degrees of categoricity of treeable degrees* (Csimá, Rossegger), and *Scott complexity of countable structures* (Alvir, Greenberg, Harrison-Trainor, Turetsky) but the argument does not have much else in common. The other two AI solutions separately encode the $\mathbb{F}_2$ -structure and make the automorphisms hard to compute; this solution does both at the same time, and as a result the $\mathbb{F}_2$ -structure is not rigid under self-embeddings. This makes the structure more compact, but also makes the verification more complicated. This solution also uses more of the language of vector spaces (e.g., quotients) more than any of the other solutions.

Exposition. The exposition is mostly appropriately concise, verifying what needs to be verified and stating without argument more obvious facts. There is some confusing terminology, e.g., “equality classes”, the “vector shift”, and so on, which one can make sense of but makes the argument harder to parse. One paragraph in particular was quite hard to parse (see comments below).

2 Recommendation

Publishable with minor revisions. The solution is correct and there are essentially only expository concerns.

3 Submission with inline comments

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Conversely, for each $c \in V$, the map τ_c which fixes I pointwise and sends $x \in S$ to $x + c$ preserves all differences $x + y$, hence preserves R . Thus

$$\text{Aut}(\mathcal{A}) = \{\tau_c : c \in V\}.$$

This already implies computable AUT-countability, with no cone oracle. Let $\mathcal{B} \cong \mathcal{A}$, and let $\sigma \in \text{Aut}(\mathcal{B})$. Transporting σ to $\mathcal{A}$, it is some translation τ_c . Choose an element i of the I -sort of $\mathcal{B}$ corresponding to some index with $D_i = \{c\}$. A Turing program with oracle the atomic diagram of $\mathcal{B}$ and with this natural number i hardwired computes σ : it queries the sort predicate, outputs n on inputs n in the I -sort, and on inputs x in the S -sort it searches for the unique y with $R(i, x, y)$. Hence every automorphism of every copy is computable relative to the copy.

It remains to prove the promised failure of Σ_1^{in} -definability. Fix a finite parameter set $P \subseteq A$. Let

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We shall show that the graph-pair (x, y) is not isolated inside $\text{graph}(\pi)$ by any existential formula over Q . This suffices, because if a countable disjunction of existential formulas defined exactly the graph, then any disjunct true of (x, y) would itself have to be contained in the graph.

Reviewer 2

“Isolated inside” means something different to me than what is happening in the proof, but I can also figure out what the proof means without too much trouble.

Let

$$\exists \bar{z} \theta(\bar{q}, u, v, \bar{z})$$

be an existential formula with parameters $\bar{q}$ from Q , and suppose it is true at $(u, v) = (x, y)$. Choose witnesses. Passing to one true conjunction of literals in a disjunctive normal form of the quantifier free part, it is enough to preserve the truth values of the finitely many literals in that conjunction.

Call an I -sort value *low* if it is at most M , and *high* otherwise. All low markers have their associated finite sets D_i contained in W . Let F be the finite set of all S -sort elements occurring among x, y , the parameters, and the chosen witnesses, and put $C = y + W$. We shall add a vector t precisely to those members of F which lie in C . The coset C contains no parameter and does not contain x .

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Reviewer 2

What is an equality class? This paragraph is very hard to parse but one can make sense of it. It is incorrect, for example, to call a pair of elements a positive or negative R -literal as an R -literal is usually a formula, but one can figure out what is meant.

Now perform the vector shift by this t . For each high equality class μ , let A_μ be the finite set of new differences required by the positive R -literals with first argument in μ , and let B_μ be the finite set of new differences forbidden by the negative such literals. By the choice of t , $A_\mu \cap B_\mu = \emptyset$.

Replace the high I -sort witnesses by new indices, all above M and in the same equality and order pattern as before, with associated sets exactly A_μ . This is possible because each finite subset of V occurs cofinally often among the D_i 's. Sort predicates, equalities, and $<$ -literals are preserved by construction. Literals involving R with an argument in the wrong sort remain false, low-marker R -literals were handled above, and high-marker R -literals are made true or false according as their resulting difference lies in A_μ or not.

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Since P was arbitrary, the required property holds for every finite parameter set. □